FED. SUP CLASS
5310

REQUIREMENTS:

SELF-LOCKING NUTS SHALL BE SUBJECT TO THE FOLLOWING LIMITATIONS:

1. SELF-LOCKING NUTS SHALL NOT BE USED AS FOLLOWS:

   a. AT JOINTS IN CONTROL SYSTEMS AT SINGLE ATTACHMENTS OR WHERE LOSS OF THE BOLT WOULD AFFECT SAFETY OF FLIGHT UNLESS THE THREADED PARTS ARE HELD BY A POSITIVE LOCKING DEVICE THAT REQUIRES SHEARING OR RUPTURE OF MATERIALS BEFORE TORSIONAL LOADS WOULD RELIEVE THE INITIAL STRESSES OF THE ASSEMBLY.

   b. ON ANY EXTERNALLY THREADED PART THAT SERVES AS AN AXIS OF ROTATION FOR ANOTHER PART UNLESS THERE ARE NO POSSIBLE TORSIONAL LOADS WHICH CAN BE APPLIED TO EITHER THE EXTERNALLY OR INTERNALLY THREADED PART IN SUCH A MANNER AS TO RELIEVE THE INITIAL STRESSES OF THE ASSEMBLY, OR UNLESS THE THREADED PARTS ARE HELD BY A POSITIVE LOCKING DEVICE THAT REQUIRES SHEARING OR RUPTURE OF MATERIAL BEFORE TORSIONAL LOADS WOULD RELIEVE THE INITIAL STRESSES OF THE ASSEMBLY.

   EXAMPLE: PULLEYS, CRANKS, LEVERS, LINKAGES, HINGE PINS, AND CAM FOLLOWERS.

   c. WITH BOLTS OR SCREWS ON JET ENGINE AIRCRAFT IN LOCATIONS WHERE THE LOOSE NUT, BOLT OR SCREW COULD FALL OR BE DRAWN INTO THE ENGINE AIR INTAKE DUCT.

   d. WITH BOLTS, SCREWS, OR STUDS TO ATTACH ACCESS PANELS, DOORS OR TO ASSEMBLE ANY PARTS THAT ARE ROUTINELY DISASSEMBLED PRIOR TO OR AFTER EACH FLIGHT.

2. BOLTS, STUDS OR SCREWS MUST EXTEND THRU THE SELF-LOCKING NUT FOR A LENGTH EQUIVALENT OF TWO THREADED PITCHES. THIS LENGTH INCLUDES THE CHAMFER.

3. SELF-LOCKING NUTS WHICH ARE ATTACHED TO THE STRUCTURE SHALL BE ATTACHED IN A POSITIVE MANNER TO ELIMINATE THE POSSIBILITY OF THEIR ROTATION OR MISALINEMENT WHEN TIGHTENING IS TO BE ACCOMPLISHED BY ROTATING THE BOLTS OR SCREWS. THE MANNER OF ATTACHMENT SHALL PERMIT REMOVAL WITHOUT INJURY TO THE STRUCTURE AND PERMIT REPLACEMENT OF THE NUTS. WHEN PROJECTION SPOT-WELDING IS USED FOR ATTACHING PLATE NUTS, CONTROL SHALL BE MAINTAINED IN ORDER THAT REMOVAL, BY DRILLING OUT THE WELDS PERMIT REPLACEMENT WITH DRILLED PLATE NUTS.

4. SELF-LOCKING NUTS THAT HAVE BEEN REWORKED OR REPROCESSED SHALL NOT BE USED BY CONTRACTORS OR FIELD MAINTENANCE PERSONNEL OF THE SERVICES.

APPROVED 15 MAR 57 REVISED (D) 30 NOV 73

(D) ENTIRE STANDARD REVISED

See Also AND10068

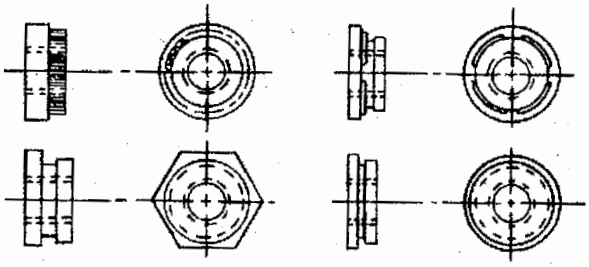
P.A. NAVY - AS Other Cust USAF-11 ARMY-AV	TITLE NUTS, SELF-LOCKING, AIRCRAFT, DESIGN AND USAGE LIMITATIONS OF	MILITARY STANDARD MS35588
PROCUREMENT SPECIFICATION NONE	SUPERSEDES AND10068	SHEET 1 OF 2

Downloaded from https://assist.dla.mil/ on 2026-10-23
Check the source to verify that this is the current version

5. THE FOLLOWING NUT TYPES, AS DESCRIBED AND ILLUSTRATED BELOW, ARE NOT ACCEPTABLE FOR USE IN AIRCRAFT STRUCTURAL APPLICATIONS. THEY MAY BE USED ON AIRCRAFT EQUIPMENT AND COMPONENT PARTS SUCH AS INSTRUMENT MOUNTINGS AND ELECTRICAL EQUIPMENT.

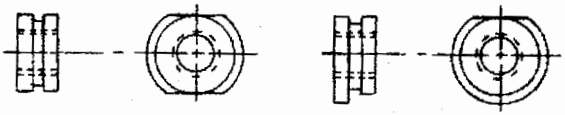
a. THOSE WHICH DEPEND ON FRICTION FOR THEIR ANCHORAGE AND TORSIONAL RIGIDITY SUCH AS SINGLE RIVET PLATE NUTS AND CLIP NUTS.

b. THOSE AS SHOWN BELOW WHICH REQUIRE EMBEDDING THE NON-APPROVED SHAPES OF POLYGONS, TEETH, KNURLES, ETC., FOR THEIR ANCHORAGE AND TORSIONAL RIGIDITY.



NOT ACCEPTABLE

6. STANDARD OR APPROVED SELF LOCKING NUTS WHICH EMBED THE SHAPES BELOW ARE ACCEPTABLE FOR USE IN AIRCRAFT STRUCTURAL APPLICATIONS.



ACCEPTABLE

7. CADMIUM PLATED SELF-LOCKING NUTS SHALL NOT BE USED IN CONTACT WITH TITANIUM AND TITANIUM ALLOY BOLTS, SCREWS OR STUDS AND IN APPLICATION WHERE THE OPERATING TEMPERATURES EXCEED 450°F.

8. SILVER PLATED SELF-LOCKING NUTS SHALL NOT BE USED IN CONTACT WITH TITANIUM AND TITANIUM ALLOY BOLTS, SCREWS OR STUDS IN APPLICATIONS WHERE THE OPERATING TEMPERATURES EXCEED 600°F.

9. SILVER PLATED SELF-LOCKING NUTS SHALL NOT BE USED WITH SILVER PLATED BOLTS.

APPROVED 15 MAR 57 REVISED (D) FOR CHANGES SEE SHEETS 1 AND 2.

Source: https://assist.dla.mil - Downloaded: 2026-10-28
Check the source to verify that this is the current version